MIL-G-17816C(YD)
10 February 1983
SUPERSEDING
MIL-G-17816B(YD)
22 July 1974

MILITARY SPECIFICATION

GAGE, VACUUM, MERCURY COLUMN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mercury column vacuum gage.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures For Packaging and Packing of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Mercury column vacuum gages shall consist of a mounting plate or case, a vertical glass or plastic tube, a mercury cistern (reservoir), a visible-type trap (catchall), a graduated scale, valves, tubing, mercury, and other components required for satisfactory operation, all assembled and ready for use when filled to the proper level with mercury.

3.2 First article. When specified (see 6.2), the contractor shall furnish one gage for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The gage shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the gage being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.5 Construction.

3.5.1 Gage. The gage shall indicate directly the vacuum in inches of mercury or when specified (see 6.2), millimeters (mm) of mercury. The gage shall be equipped with a valve or a means through which the mercury may be drained from the cistern (reservoir). A means to set the scale at zero without disconnecting the gage shall be provided. The maximum error of the gage shall not exceed 0.05 inch mercury over the full range of the scale when tested in accordance with 4.5.

3.5.2 Case. When a case is provided, the case shall have a glass or clear plastic front, so installed as to cause a dusttight fit. Gages shall be designed for wall, table, panel, or a combination mounting, as specified (see 6.2). The cases shall be provided with the necessary fittings for mounting.

3.5.3 Tube and mercury cistern. The indicating tube shall be clear glass or plastic. The tube may have a white strip in back of the mercury bore to provide a better reading contrast. The tube bore shall be not less that 0.10 inch, and the tube shall be mounted on a mounting plate or case (see 3.5.2). The lower end shall be immersed in or connected to a mercury cistern, and both shall be protected against the entrance of moisture, dirt, or other foreign matter. Each gage shall be provided with a charge of mercury sufficient to fill the cistern and the indicating tube to the proper level when the cistern is under atmospheric pressure and the scale is properly adjusted.

3.5.4 Trap. When specified (see 6.2), a trap (catchall) shall be provided for visibly showing trapped condensate and shall prevent any condensate or dirt from being drawn into the mercury tube. The trap shall be made of cast iron, steel, or other suitable material and shall have a drain connection. The drain connection, when tightened, shall not leak.

3.5.5 Scale. A calibrated scale shall be furnished which shall indicate from 0 to at least 30 inches of mercury. The scale shall be subdivided into 0.1-inch graduations and shall have figures at 1-inch intervals. The 0.5-inch graduations shall be made visibly longer than the others. The graduations shall be permanently and accurately marked and shall have clean, sharp edges. Either the scale or the mercury cistern shall be provided with a scale-zero adjustment mechanism which shall be operated from the outside of the case. The scale or mercury cistern and all units shall move without binding or sticking. When specified (see 6.2), the scale shall be calibrated in mm having subdivisions every 2 mm and figures every 50 mm.

3.6 Piping. Unless otherwise specified (see 6.2), piping shall be the manufacturers standard materials with the necessary fittings. The vacuum connection of the trap shall be not less than 0.125-inch nominal pipe size. The vacuum connection side shall have standard pipe tapered threads unless otherwise specified (see 6.2). Screw threads shall be in conformance with FED-STD-H28. All piping, fittings, connections, and valves shall be tight and leakproof.

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3.7 Interchangeability. All gages of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.8 Treatment and painting Unless otherwise specified (see 6.2), the gage shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the gage other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.9 Identification marking. Identification shall be permanently and legibly marked directly on the gage or on a corrosion-resisting metal plate securely attached to the gage at the source of manufacture. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.10 Workmanship. The quality of workmanship shall be such as to produce gages that are in accordance with the requirements of this specification and that are so constructed as to insure proper functioning of all parts of the instrument.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete gage when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All gages offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level S-2 and an acceptable quality level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 2.5 percent defective.

4.4 Examination. Each sample selected shall be examined for defects listed in table I.

TABLE I. Classification of defects.

Classification	Defects	Requirement paragraph
Critical	None defined	
Major:		
101	No provision for draining mercury	3.5.1
102	Fittings for mounting gage missing	3.5.2
103	Charge of mercury missing	3.5.3
104	Scale not legible	3.5.5
105	Scale zero adjustment failure	3.5.5
106	Pipe and fittings leaking	3.6
107	Component parts not interchangeable	3.7
Minor:		
201	Trap drain connection leaks	3.5.4
202	No corrosion-resistant treatment	3.8
203	Identification marking not as specified	3.9

4.5 Tests. Each gage selected in accordance with 4.3.2 shall be tested and any unit failing to pass the following tests, as applicable, shall be rejected.

4.5.1 Calibration test. The gage shall be subjected to calibration tests at the cardinal points over the specified range by comparison with a standard absolute pressure mercury gage or precision manometer. The calibration shall show no deviation greater than 0.05-inch mercury from the correct vacuum at the point tested.

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4.5.2 Leak test. The gage shall be subjected to 5 half-hour periods of 27-inch mercury vacuum alternated with half-hour periods of atmospheric pressure, during and after, which shall show no deviation in excess of 0.05-inch mercury from the correct vacuum.

4.6 Packaging inspection. The preservation, packing, palletization and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Gages. The gages shall be individually preserved by method III in accordance with MIL-P-116. If the gage is provided with a carrying case, the case may be utilized as the initial container for the preservation method. Contents of the carrying case shall be cushioned and protected in accordance with the applicable requirements of MIL-STD-1186.

5.1.1.2 Mercury. If not included within the gage, sufficient mercury for each gage shall be individually packaged.

5.1.1.3 Consolidated packaging. Each gage shall be packaged in a fiberboard box conforming to PPP-B-636, class weather-resistant. Sufficient mercury for each gage shall be packaged with the gage. The gage and mercury shall be cushioned and protected in accordance with MIL-STD-1186.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be accordance with MIL-STD-794. Containers shall be selected from table I for the appropriate level.

5.2.2 Commercial. Gages shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with applicable carrier rules and regulations.

5.3 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria is met:

a. Load shall consist of four or more unskidded containers; and,
b. Load shall utilize a minimum of 80 percent of the pallet base.

5.4 Marking. Marking shall be in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. Gages covered by this specification are intended for use as indicators of vacuum, up to 30 inches of mercury.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of document required if other than as specified (see 2.1.1).
c. If first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When scale shall indicate vacuum in mm of mercury (see 3.5.1 and 3.5.5).
e. Whether gages shall be designed for wall, table, panel, or combination mounting (see 3.5.2).
f. If trap is required (see 3.5.4).
g. When piping, fittings, and screw threads are other than as specified (see 3.6).
h. Treatment and painting if other than specified (see 3.8).
i. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a Contract Data Requirements List (DD Form 1423), and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one gage. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Review activity:

DLA - GS

Preparing activity:

Navy - YD

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MIL-G-17816C(YD) - GAGE, VACUUM, MERCURY COLUMN

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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Replaces edition of 1 Jan 72 which may be used.

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